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Inside and Outside the Online Study Room: An Exploratory Study of Student Practices and Perceptions

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Abstract. The increasing use of online study rooms raises critical questions regarding the dynamics and challenges of virtual learning environments. This study explores students' experiences in remote study settings through semi-structured interviews with 13 university students. We explore the factors influencing their participation, the obstacles they face before, during, and after study sessions, and the role of social interaction in sustaining motivation. Our findings highlight the significance of the perceived sense of responsibility enhanced by camera usage in maintaining concentration and engagement. Moreover, feelings of intimacy and belonging, particularly when studying with close peers, play significant roles in motivation and focus. Students also reported challenges, such as coordinating schedules and managing distractions in group study sessions. Based on these insights, we propose design implications for enhancing online study environments. We emphasize fostering a stronger sense of community, minimizing distractions, and facilitating effective collaboration. Our contributions inform the design of more inclusive and engaging virtual study platforms, with broader implications for learning communities and online collaboration tools.

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Introduction

Video-based communication is a ubiquitous method of interaction in diverse contexts. Many individuals rely on video calls for emotional support and close connection with their families (Brush et al., 2008; Cao et al., 2010). They also use video-mediated platforms to socialize, share meals, watch movies, or play games together (Heshmat and Neustaedter, 2021; Bryld et al., 2020; Brubaker et al., 2012).

The widespread adoption of video-mediated platforms reshaped education and a shift accelerated by the COVID-19 pandemic. Virtual classrooms on platforms like Google Classroom facilitate remote learning[1], while high-quality courses on Udemy[2] and edX[3] provide accessible educational content. However, the absence of face-to-face interactions often diminishes students' motivation and focus. To address this, many students watch or create "Study with Me" videos on YouTube for their motivations (Jhuang et al., 2022; Lee et al., 2021; Kim and Ryoo, 2024), while others seek study partners through online communities and virtual study rooms such as StudyVerse[4], StudyStream[5], and Focusmate[6].

HCI and CSCW research extensively examined students' experiences in virtual classrooms. Prior research explored perceptions and usage patterns of online study environments, while Dyer and Mandernach (2024) proposed methods to support learners with disabilities in online education. Several researchers also introduced tools and guidelines for enhancing remote learning. Liu et al. (2024) demonstrated how large language models(LLMs) can power virtual agents to foster active student participation in VR-based classrooms. Labrie et al. (2022) interviewed university instructors, identifying four key guidelines to mitigate technical challenges in videoconferencing-based education. Furthermore, Na et al. (2023) introduced Students' Understanding Visualizer, a tool that enables instructors to assess real-time student comprehension during online lectures.

Despite the growing area of virtual study platforms, little is known about why students study together without instructors, or how they form and sustain their online study groups. This informs our two main research questions: **(1)** What is the lifecycle of online study rooms for students? **(2)** How can we enhance student experience in these online study environments?

We examine students' experiences studying remotely with peers to address these questions. Through semi-structured interviews with 13 university students, we investigate their motivations, practices, and challenges in an online study setting. By analyzing these experiences, we identify key difficulties and propose design implications to enhance student engagement and support community

[1]https://edu.google.com/intl/ALL_us/workspace-for-education/products/classroom
[2]https://www.udemy.com
[3]https://www.edx.org
[4]https://csw.live/study-verse
[5]https://www.studystream.live
[6]https://www.focusmate.com

managers in fostering more effective online study environments. The main contributions of this study are as follows.

- We provide an in-depth understanding of how students perceive and engage in remote online study.
- We highlight critical obstacles students face before, during, and after virtual study sessions.
- We offer recommendations to help students and community managers create more engaging and sustainable online study experiences.

Related Work

This section reviews prior studies on students' experiences in social learning environments, including their peer-based and self-regulated learning engagements. We also examine research on remote study practices, including the role of digital platforms and video-based study communities. We then present the limitations of existing studies and discuss the need for further research to support students in remote study rooms.

Understanding Students in Social Learning Environments

Peer-Based and Self-Regulated Learning

Researchers studied how students interact in group studies, exploring their motivations, needs, and challenges when studying with peers. Studies employing interviews and surveys examined what drives students to engage in group studies (Hendry et al., 2005; Bradshaw and Hendry, 2006; Dolmans and Schmidt, 2006) and the challenges they encounter in such settings (Cho et al., 2023a). Recent work in CSCW further highlighted that the presence of peers can act as a subtle yet effective form of self-regulation by providing ambient social pressure, thereby enhancing concentration, even in virtual contexts (Cho et al., 2023b). For example, Cho et al. (2023a) investigated students' hiding and revealing their needs during an online group-study session. Additionally, Fang et al. (2022) suggested a tool to support students in note-taking in individual and social learning environments. Moreover, recent studies demonstrated that integrating synchronous interaction tools into study environments can foster a stronger sense of belonging and accountability, thereby positively affecting self-regulated learning strategies (Won et al., 2021; Peacock et al., 2020).

Studies also examined peer learning within specific academic disciplines, such as medicine (Ali, 2016; Lovell, 2015) and STEM (Sandoval-Lucero et al., 2012). For instance, Lovell (2015) reported that social identity within study groups enhances group cohesion, which, in turn, promotes academic engagement. Ali (2016) explored how medical students use Facebook for social learning. Moreover, Sandoval-Lucero et al. (2012) conducted a focus group study to understand the experiences of STEM students participating in self-initiated groups.

“Study with Me” communities

Outside of traditional peer learning, students also engage in self-regulated study using “Study with Me” videos, recordings, or live streams of individuals studying alone. These videos captured students’ natural study behaviors, providing a sense of companionship and motivation for others to study remotely. Recent research presented that such videos not only help viewers recreate the ambient pressure of a physical study space but also allow the customization of study environments to suit individual needs (Kim and Ryoo, 2024; Lee et al., 2021). Researchers investigated students’ motivations and practices in using “Study With Me” videos (Kim and Ryoo, 2024; Jhuang et al., 2022; Lee et al., 2021; Wang and Zhang, 2021). For example, Lee et al. (2021) found that viewers actively select videos that provide the desired level of ambient presence, balancing too much distraction with insufficient peer pressure.

Examining Remote Work Models and Challenges

While many studies focused on remote study environments, others examined remote work experience and proposed design improvements for virtual collaboration. Although these studies primarily addressed professional settings, insights into remote work practices (such as maintaining engagement and managing distractions) can inform the design of virtual study spaces. Schuh et al. (2024) taxonomized the remote work concept in organizations. They suggested that this classification enables organizations to efficiently design lasting and integrated remote work environments. Borse et al. (2022) conducted surveys and focus group studies, investigating employees’ motivation and productivity factors in remote working, while Kabir et al. (2025) focused on care providers to understand their challenges. Although these studies focus on remote work, the underlying themes of productivity, ambient presence, and self-monitoring also apply to online study settings, further emphasizing the importance of social presence in digital environments.

Gaps in Prior Studies

Prior studies showed that students’ experiences and social perceptions of remote study rooms remain underexamined. Notably, while much research investigated the technological aspects of video conferencing tools and remote work, few studies have critically examined the nuanced dynamics of virtual study spaces, especially regarding how ambient peer presence and personalized study environments influence self-regulated learning. Thus, further research is necessary to understand how students interact with each other during and after a study session. We aim to extend the existing literature by exploring both in-session behaviors and their subsequent effects on motivation and focus, thereby addressing these identified gaps. With a better understanding of students’ experiences in remote study environments, we propose design opportunities for technology to support them.

P#	*Gender*	*Age*	*Major*
P1	Female	22	Theatre
P2	Male	23	Advanced Materials Science and Engineering
P3	Female	21	Integrative Biotechnology
P4	Female	23	History
P5	Female	23	Immersive Media Engineering
P6	Female	20	Statistics
P7	Female	23	Computing and Informatics
P8	Male	23	Media and Communication
P9	Female	20	Chemistry
P10	Female	22	Business Administration
P11	Male	21	Statistics
P12	Female	28	Applied Artificial Intelligence
P13	Female	23	Business Administration

Table I. The demographic information of the interview participants..

Method

Participants

To explore the students' experiences and challenges in online study rooms, we conducted semi-structured interviews with 13 participants (see Table I). Participants were recruited from university communities in South Korea based on the following inclusion criteria: (1) current university students, (2) active participation in online study rooms within the last six months, and (3) aged between 18 and 64 years.

A total of 28 individuals registered in the study. However, 15 were excluded because of infrequent participation in online study rooms (e.g., once every six months). Ultimately, 13 students participated in the interviews. The participants ranged from 20 to 28 years, with an average age of 22.5 (*SD*=2.0). Eleven were undergraduates, one was a master's student (P5) and the other was a Ph.D. student (P12). Their academic backgrounds varied, with eight majoring in STEM, four in business and social sciences, and one in theatre.

Interview Study

All interviews were conducted remotely over Google Meet[7] for 35 minutes to 1 hour. The participants provided verbal consent to record the interviews and anonymous data. The participants acknowledged that their participation was voluntary. Each participant received approximately 7 euros (10,000 KRW) of cash to complete the study. All interviews were recorded, and the researchers transcribed them after the study. Once the data analysis was completed, we deleted all recordings from the database. The interviews included the following topics.

[7]https://meet.google.com

1. Engagement and Interactions: *Right after joining the online study session, are there any aspects that catch your attention(e.g., your face, background, microphone, camera angle, number of participants)? When connected to an online study session, do you check the screens of other participants during your study? What is the frequency and reason for doing so?*

2. Challenges in the Online Study Session: *In the study preparation phase, what inconveniences did you encounter when using tools to gather participants? How do these factors affect the preparation process? What difficulties did you face during the study session while studying or using the features of the tools? What aspects did you find inconvenient? What inconveniences did you experience in the poststudy review phase? And what difficulties or inconveniences did you encounter with the tools you used?*

We gathered statements from the interview transcripts to analyze the qualitative data. The transcripts were coded separately. Then, to identify key themes, insights, and patterns that repeatedly appeared in the transcripts, researchers conducted affinity diagramming on Figma[8] and Notion[9]. We identified notable themes related to why and how students participated in online study rooms, and the challenges they faced during the study session. The user study was conducted with approval from the Institutional Review Board at Sungkyunkwan University in South Korea (SKKU 2024-11-032-004).

Results

We classify our key findings into two primary categories: internal and external. Internal factors include the elements within the study session and configurations that participants can modify, such as camera usage, peers' screen displays, sound and microphone, filters, and other interactive features. External factors include aspects outside the session, including pre- and post-session arrangements, fostering intimacy and belonging, facilitating interaction and feedback, and promoting social responsibility.

Internal factors of online study rooms

We found that participants engaged with features in video-conferencing tools and with peers during the online study session. As shown in Table II, the internal factors influencing online study sessions include various aspects such as camera usage, screen-related distractions, and microphone settings.

[8]https://www.figma.com
[9]https://www.notion.com

Internal Factors in Online Study Sessions		
Factor	**Subcategory**	**Participants (N)**
Camera turned on	Self-surveillance	11
Screen/others	Competitive spirit	12
	Peers' movement distraction	3
	Peers not paying attention	7
	Peers' camera off	6
Screen/my own	Privacy concern	2
	Low self-awareness	11
Microphone	Presence of others	6
	Distraction	8

Table II. Internal factors in online study sessions.

Camera usage

Participants noted that camera usage was one of the most essential features of motivation. Eleven participants mentioned that keeping their cameras on and showing their appearance to others improved concentration. For example, P12 mentioned that awareness of others in her study session created tension.

"*I know if I don't keep my hands on the desk, I'll probably start doing something else, so I try to position the camera to capture my desk with wide-angle view.*"(P12)

In addition, P7 explained that keeping the camera on made her feel a sense of self-surveillance:

"*By keeping the camera on, I felt I couldn't leave my spot and was more accountable for focusing.*"(P7)

Furthermore, most participants adjusted the angle to ensure that their upper bodies and materials were visible. This allowed them to monitor their behavior better, positively impacting their focus. For example, P2 explained that she and her friends set rules to ensure that their faces, upper bodies, and desks were visible to the camera. P4 also mentioned that she felt distracted when her hands were not visible.

"*If my hands aren't captured, I know I'll get distracted, so I make sure my desk is in a wide-angle view.*"(P4)

Screens - Peers' Screens: Motivation and Distraction

Watching peer studies during the session motivated the participants by fostering a sense of competitive spirit (Cho et al., 2023b), which inspired them to push themselves harder. Some participants also monitored their peers' screens and gave verbal reminders when others appeared distracted. For example, P1 said, "I noticed my friend getting distracted, so I asked, 'Why aren't you studying?' and encouraged her by saying, 'Let's get back to work'."

However, peer screens also cause distractions. Participants noted that movements on peers' screens, camera-off screens, or inattentive behavior reduced

focus. P1 found it distracting when even one friend seemed disengaged, especially if another had their cameras off.

"*When I kept the screen open while studying, the movement on it caught my attention and became distracting.*"(P2)

Self Screens - Privacy Concerns and Self-Awareness: In contrast, eleven participants reported low privacy concerns or self-awareness during online study sessions. They mainly checked their screens for technical reasons and were largely unaware of their presence. For example, P9 said, "I don't check my screen except when setting it up initially," and P3 stated, "I only check occasionally to make sure my camera is on, but I don't care about my facial expression or posture." P8 said that while she frequently checked peers' screens at the beginning, but did not recheck her screen when setting it up.

This sense of ease and reduced self-awareness appears to be associated with studying with familiar peers. In contrast, some participants expressed discomfort when studying with strangers and mentioned their privacy concerns. P2 said, "*Since I usually connect from home, a certain amount of my private space is exposed, and I don't want to share it with strangers.*

Sound and microphone

Participants reported that sounds influenced their focus during the online study sessions. Eight participants preferred to mute their microphones to stay focused. They were especially mindful of avoiding noise that could distract them from the group. For instance, P2 said that muting was a rule to prevent sound from interfering with the concentration.

By contrast, six participants preferred to keep their microphones on. They said that background sounds like typing or page flipping enhanced the sense of studying together and made communication feel more natural. P13 found these sounds motivating, as they indicated that their peers were also working.

"*It was nice to hear peers working harder, and whenever I had something I didn't know, I could ask and get an answer ... We kept our microphones on so we could hear typing sounds and know that everyone was studying together.*" (P13)

Similarly, P12 noted that a moderate level of background noise, if not too loud, helped maintain focus and fostered a shared atmosphere. *"With just the right amount of white noise, I found it easier to focus. Even though we weren't physically together, it felt like we were studying as a group, so most of us kept our microphones on."*

Tools and Features

We found that various interactive features, such as hand raising, sound control, and emojis, facilitate communication during online study sessions.

Filters and appearance: Eight participants stated that they used filters (e.g., backgrounds or face masks) to ease tension and lighten their mood during the sessions. They mentioned that the filters were helpful for stress relief. For

example, P5 said that she used mask filters to make sessions fun when they became dull.

However, some participants found the filter distracting. P8 noted that face filters and virtual backgrounds drew attention away from studying and reduced the sense of shared presence. "*It's hard to recognize the person, and it feels like they're not in the same space. Since the character keeps appearing, I try not to use it.*"(P8)

Similarly, P12 explained, *"When other friends' face changes into a character or some cute image, I find it distracting rather than helpful for studying."*

Hand-Raising and Sound(Noise) Control: The hand-raising feature allows participants to communicate fluently with their peers. Participants often used the hand-raising feature before speaking to avoid interrupting their peers with sudden noise. This approach mitigates noise and facilitates natural communication without surprising peers. For instance, P11 reported using a handraising function before turning on the microphone.

"*Becuase we're all studying, turning on the microphone suddenly feels disruptive. Therefore, I use the hand-raising feature to signal that I have something to say. If it's okay, I speak right away, or if not, I bring it up during the break.*"(P11)

Emojis and reactions: Emojis and reactions provided another way to communicate during the session without interruptions. Emojis allowed the participants to express their emotions while providing contextual feedback. For instance, P8 noted that they frequently employed reactions and emojis to express their feelings, without compromising their focus.

Chatting: Chatting enabled effective communication without interrupting the study process. Participants found that text-based tools allowed them to exchange questions and feedback without distractions. P9 mentioned that they used personal messages to quickly address questions, which helped them maintain their focus on the study session.

Time limit: Finally, participants reported that setting a time limit for study sessions contributed to maintaining concentration. Some participants used the built-in time limit feature in Gooroomee[10] to structure sessions and schedule regular breaks. Using tools such as Google Meet or Zoom, others established their own time-limit rules to manage study session productivity. For example, P3 took advantage of the automatic cutoff every 40-50 minutes to concentrate during the zoom study session.

Other factors

In addition, participants mentioned that accessing an online study session was an important motivating factor. Upon entering the study room, participants reported feeling an immediate obligation to concentrate, which increased their motivation to participate. For instance, P9 noted that mere awareness of participating in an online study facilitated their focus and motivation.

[10]https://gooroomee.com?lang=en

External Factors in Online Study Sessions		
Factor	**Subcategory**	**Participants (N)**
Pre-study session	Coordinate schedules	6
	Setting rules	9
	Prepare before entering	4
Intimacy	Common goal	5
	Sense of intimacy	7
	Sense of belonging	4
	Drawbacks with close friends	3
Interaction	Encouraging	7
	Conversation	2
Post-study session	Feedback and review each other	5
	Coordinate schedule	3
	Encouraging each other	3

Table III. External factors in online study sessions..

External factors of online study rooms

Participants also interacted outside the sessions for scheduling and social coordination. We found that strong social bonds among members helped sustain participation in online study sessions. As shown in Table III, the external factors influencing online study sessions included pre-study session preparation, intimacy and belonging, and interaction and feedback, with varying numbers of participants reporting each factor.

Pre-study session

Six participants scheduled the session through various tools, such as KakaoTalk's[11] voting feature, Notion calendar, and When2meet[12]. However, scheduling through group chats is often inefficient, as not everyone responds promptly. Using When2meet or voting is time-consuming, requiring everyone is input, which makes scheduling cumbersome and inefficient. For instance, P9 noted that adjusting the schedule from group chats to chats was inconvenient. Scheduling through group chats was challenging because not all participants could respond simultaneously.

In contrast, nine participants said that setting clear rules beforehand helped maintain focus. They agreed to turn on their cameras and adjust their angles to keep the study activities visible. For instance, P2 explained that she and her peers ensured that their faces and upper bodies were shown, and took breaks at agreed intervals. In addition, they employed a method of focused study interspersed with designated breaks.

[11]https://www.kakaocorp.com/page/service/service/KakaoTalk
[12]https://when2meet.com

Intimacy and belonging

Ten participants indicated that they experienced an enhanced sense of belonging and responsibility when engaging in social interactions with peers. For instance, P7 said she could concentrate better when studying with her friends because of the comfortable atmosphere compared with studying with strangers.

Conversely, participants indicated that studying with strangers was challenging because of the burden of personal space exposure and lack of social connection. Unfamiliarity reduced their sense of accountability and engagement, making it harder for them to stay committed. P4 felt overwhelmed when studying an unknown individual. Similarly, P11 remarked, 'When studying with strangers, there's no connection, so it feels like it does not matter whether I stay or leave,' and noted that coordinating with unfamiliar members was also difficult. Because of these concerns, some participants avoided joining online study sessions with strangers.

Interaction and feedback

Participants found that brief conversations with peers during breaks helped them to recharge and refocus. It also enabled higher concentration levels and more efficient learning. For instance, P1 reported feeling more refreshed after engaging in dialogue with colleagues, which enabled them to take a more focused approach to their academic endeavors. Similarly, P4 mentioned, "After having a short conversation during the break, I could focus again," and P3 shared that discussing the study topic with a friend helped her refocus.

Moreover, participants mentioned mutual support and encouragement during and after the study sessions. They noted that recognizing and encouraging each other's efforts was a significant factor in maintaining a positive attitude toward learning. P13 indicated that close friends offered encouragement if they struggled to concentrate during the study.

However, some participants noted that when conversations became too frequent or extended beyond short breaks, they disrupted the study atmosphere and made it harder to refocus. P1 added that extended chatting with close friends sometimes made it harder to concentrate, leading to time loss.

Social responsibility

We found that committing to a study schedule with friends and accessing the study to meet the schedule itself gave a sense of belonging and compulsion, and played an important role in the rates of online study session participants. Three participants explained that this coercion engenders an environment where they exercise control and focus on distractions by making them self-responsible for their study time. For instance, P10 articulated that the obligation to commit and engage in study enabled him to prioritize moments when she could not study freely.

"*When I study alone, I often say, "Let's just postpone it until tomorrow." If I ask my friends to study Zoom, I will be forced to study. So, I think I have no choice but to use it in that sense.*"(P10)

However, some participants noted that not all group members were equally committed, and uneven levels of motivation sometimes disrupted the overall focus or atmosphere of the study session. In addition, the participants became more active in online studies during the exam period. Five participants argued that the efficacy of online studies is amplified when shared objectives are present, such as during a designated test period with a clearly defined deadline. They explained that when peers share a common goal, they feel more connected with each other. This feeling helps them work better. P4 provides an example of this idea. When people share the same goal, they feel that they belong. This encouraged them to encourage each other and participate more in the study.

"*I still remember those nights when my friends and I stayed up studying together on Google Meet during exams.*"(P4)

Post-study session

After the study session, participants interacted with peers by giving feedback, coordinating the next session, and sharing encouraging words, which helped boost motivation. Some suggested that writing and sharing study diaries could help them provide feedback and learn from each other. In addition, it was said that sharing motivational statements after the study ended positively affected participants' motivation.

Some participants preferred to schedule the next session immediately after the study ended, often using group chats or comment pages in real-time. They noted that scheduling immediately after the session helped to maintain continuity and improve efficiency. For instance, P6 indicated that the subsequent study would be conducted in the group chat room immediately following its conclusion.

Other factors

Other factors included study duration, group size, and key elements that influenced online study. First, although studying online varied, four participants reported spending more than two hours in the study sessions. P11 added that he always finished within two hours because he lost concentration after more than two hours.

Second, four participants agreed that groups of three to four would be sufficient for the study session. They explained that a group comprising three or more individuals fostered optimal motivation and generated an organic study atmosphere. Conversely, a one-on-one study environment disrupts concentration, and groups of two or fewer individuals are suboptimal. P4 also noted that a one-on-one study hindered his ability to concentrate.

Discussion

Studying with close peers fosters belonging and focus, although it also presents distractions that require a balance. To provide an overview, Table IV summarizes the advantages and challenges of the online study session. In this section, we explore how relationships between peers impact motivation in online studies, describe particular challenges students face in online studies, and present design implications for technology to address these challenges.

Visual and Auditory Presence in Supporting Focus and Engagement

Camera Use for Accountability and Social Engagement

All participants reported preferring to keep their cameras during online study sessions, as doing so supported sustained focus and heightened their sense of responsibility. Being aware that their own screens were visible to others encouraged self-surveillance and reduced distractions (Cho et al., 2023b). In addition, seeing peers on the camera enhances their sense of presence and creates a shared study atmosphere. Some participants reported that watching peers study diligently motivated them to stay engaged. These findings are consistent with the social presence theory (Short et al., 1976) and social learning theory (Bandura et al., 1986), which suggest that visual cues and observing peers' behavior can enhance motivation and engagement.

Auditory Presence and Shared Atmosphere

In addition to visual cues, auditory presence fosters a shared atmosphere. Participants noted that keeping microphones allowed them to hear subtle study-related sounds, such as typing or page flipping, reinforcing their sense of co-presence. Such ambient audio has been found to support focus and sense of connection in virtual settings (Jo and Jeon, 2022; Nichols et al., 2000).

Conversely, when peers turned off their cameras or remained silent, participants felt less motivated and connected, highlighting the role of visibility and audibility in sustaining group engagement (Short et al., 1976).

How Close Relationships Enhance Focus/Motivation in Online Study

This section explores how close peer connections cultivate a heightened sense of belonging and responsibility, driving a stronger focus and motivation in online study environments.

The Role of Belonging in Enhancing Focus and Motivation with Close Peers

Most participants reported experiencing a stronger sense of belonging and responsibility in online study sessions with their close peers. "Close peers" were

typically friends from the same department or long-term acquaintances. This familiarity motivated the participants to engage more actively, such as by encouraging distracted peers during sessions and reinforcing belonging.

This sense of social connection contributed to greater follow-through on study commitment. Close relationships appear to heighten responsibility in online settings, as students studying with familiar peers are more engaged in shared goals (Xu et al., 2020). Similarly, Rothstein and Pierotti (1988) noted that people are more cooperative and feel a stronger sense of duty in close relationships. Participants in our study were also more likely to stay committed to sessions with friends than with strangers. In particular, preparing for exams together, especially during exam periods, further strengthened students' sense of responsibility toward one another, increasing their motivation and focus. This aligns with Peacock et al. (2020), who states that a sense of belonging helps students stay committed to their studies and is resilient against stress and anxiety. Additionally, Won et al. (2021) highlighted that a sense of belonging influences adaptive help-seeking behaviors and plays a critical role in supporting self-regulated learning, ultimately enhancing students' focus and participation. Therefore, fostering a sense of belonging can promote engagement and concentration in learning environments.

Reduced Self-Awareness and Improved Focus with Close Peers

Cho et al. (2023b) indicated that excessive self-awareness during virtual study sessions emerged when users became overly conscious of their appearance on the video and became uncomfortable with the camera, which in turn interfered with concentration. However, our findings indicate that participants generally experienced low self-awareness during the study sessions when they were familiar with their peers.

Some participants mentioned that they rarely checked their screens or paid attention to their appearance during the session. This difference may be attributable to the study of familiar peers. Most participants primarily studied their close peers.

As Franzoi et al. (1985) suggested, individuals feel more at ease and self-disclose more freely in close relationships. By contrast, studying with unfamiliar peers may increase self-awareness and lead to discomfort.

Although research on this topic remains limited, our findings suggest that peer familiarity is crucial for shaping self-awareness in online study settings. Future studies should explore how social dynamics influences self-perception and engagement in virtual learning environments.

Light Conversations on Focus During Study Breaks

The participants' close relationships enabled casual break-time conversations, which helped reduce fatigue and restore focus. As noted earlier, interactions, such as light chat and encouragement, were especially effective in helping students recharge without disrupting their concentration.

Category	Subcategories	Details
Advantages of Online Study with Close Friends		
Intimacy and Belonging	Sense of intimacy	Familiar settings lower anxiety and boost focus.
	Sense of belonging	Builds shared responsibility and motivation.
	Low self-awareness	Less concern about appearance improves focus.
	Low privacy concerns	Sharing space with friends feels less intrusive.
	High social responsibility	Studying together increases accountability.
Interaction	Encouraging	Mutual support encourage motivation.
	Conversation	Brief talks during breaks recharge focus.
	Structured routine	Scheduled sessions build commitment.
Challenges of Online Study with Close Friends		
Scheduling & Group Composition	Coordinate schedule	Difficulty aligning schedules.
	Suboptimal group size	Too few members lower accountability and invite chatter.
Interaction Dynamics	Conversation	Off-topic talks can derail focus.
	Uneven engagement	Inconsistent participation lowers group drive.
	Overuse of interactive features	Excessive emojis or filters distracts studying.

Table IV. Advantages and challenges of online study sessions with close friends..

Prior research suggests that microbreaks, including social interactions, can restore energy and reduce fatigue (Albulescu et al., 2022). Such interactions can support cognitive recovery if well-managed, although they may also become distracting (Hossain et al., 2024). These suggest that incorporating light social interactions during breaks may contribute to focus and reducing fatigue in online study sessions.

Challenges students faced during the online study session

Scheduling Difficulties

As discussed above, scheduling study sessions often required group agreement via tools such as KakaoTalk or When2meet, which participants found to be inefficient. While studying with close peers to foster comfort, coordinating schedules still demanded effort and often became a source of friction. This suggests that, even in informal peer-based settings, difficulties in coordinating schedules may affect participants' willingness to engage consistently.

Distractions from Social Interactions

While conversations fostered a supportive atmosphere, some participants noted that it sometimes became excessive, making it difficult to concentrate and leading to time being wasted. Although the participants valued social interactions, balancing casual conversations with a focused study was necessary.

Uneven engagement is another issue. As noted in the Results, some participants observed uneven engagement within the group, with some members being highly committed and others being more passive. In some cases, less-motivated members contributed to a decline in the overall study atmosphere. Additionally, filters and virtual backgrounds are sometimes perceived as distracting, particularly during moments requiring sustained attention. Some participants noted that face filters could reduce their sense of co-presence by making it harder to recognize peers. When others' faces change into characters or playful images, this can lead to laughter and momentarily disrupt focus. Thus, while such features may lighten mood, they may hinder focus if overused.

Design Implications

Our findings suggest that students' focus, engagement, and sense of responsibility in online study environments are shaped by a combination of social and structural factorssuch as visual and auditory presence, session rules, peer awareness, and opportunities for informal interaction. While existing platforms like StudyStream and Gooroomee[13] promote participation by connecting users with shared goals, they may not fully support the nuanced routines and social cues that contribute to effective peer-based study.

To address these gaps, we outlined a set of design implications aimed at enhancing virtual study environments. These suggestions focused on fostering mutual presence, managing interaction rhythms, and supporting continuity across sessions. Rather than prescribing fixed solutions, our goal was to offer possible directions for making remote study platforms more socially attuned and structurally supportive.

Turning on the Camera Upon Entry

Consistent camera use fostered participants' sense of accountability and motivation, particularly when the sessions involved peer observation. Encouraging students to activate their cameras at the beginning of a session may help reinforce this sense of responsibility and contribute to a sustained focus throughout the session.

Camera visibility in virtual study rooms fulfills users' need for enhanced presence and self-surveillance, although it must be balanced against privacy needs (Cho et al., 2023b). This balance becomes especially important in settings where participants are unfamiliar with one another. Such practices are most effective when supported by shared norms and expectations of visibility. Platforms

[13] https://gooroomee.com/?lang=en

could facilitate this by providing gentle camera-on prompts or soft default settings along with optional visual filters or blur modes to reduce discomfort. Therefore, we recommend that online study platforms and facilitators request or require participants to activate their cameras upon entering sessions to humanize the interaction and foster immediate group trust.

Managing Microphone

Keeping microphones during the study sessions helped some participants stay focused and feel a stronger sense of presence, particularly when subtle background sounds, such as typing or page-flipping, were audible. These ambient cues appear to reinforce a shared sense of studying. However, excessive or irrelevant noise often disrupts concentration.

To address this, we suggest implementing microphone protocols that preserve the benefits of ambient audio, while minimizing disruptions. Facilitators may consider allowing moderate study-related background sounds while discouraging unrelated noise. Platforms can support this by offering ambient audio modes or subtle sound filters that enhance desirable cues and suppress distraction. Additionally, visual indicators of background presence, such as icons showing typing activity, may help maintain social awareness without relying on the voice. Managing microphone use in this manner also reflects broader challenges in collaborative systems of balancing awareness and interruption (Shen and Sun, 2002).

Using Pre-Notifications

Some participants expressed concern that speaking abruptly, such as turning on a microphone without notice, might disturb others. To avoid this, they used non-verbal signaling tools like the "raise hand" feature to signal their intention to speak[14],[15].

This approach gave others time to prepare, helping maintain a smooth study flow. Salem and Earle (2000) emphasizes the importance of non-verbal feedback in learning environments. Similarly, emojis and chat functions enable brief, unobtrusive interactions and feedback. These signals recall early media space designs in which non-verbal cues manage transitions between awareness and focused interaction without causing undue disruption (Gaver et al., 1992).

We recommend that researchers and platform designers support pre-notification practices, such as non-verbal signaling and timely reminders, to help participants prepare for and reduce unnecessary disruptions.

[14] https://support.zoom.com/hc/en/article?id=zm_kb\&sysparm_article=KB0068290

[15] https://support.google.com/meet/answer/10159750

Encouraging Session Continuity with Post-Study Scheduling Tools

Participants often coordinated the next study session immediately after the previous session, typically via chat. This spontaneous scheduling, while still in the study mindset, helped maintain momentum and reduce the friction of re-coordination. To support this behavior, platforms should offer lightweight post-session scheduling features, such as quick polls, suggested times, or smart reminders, which appear as the session wraps. Integrating these tools seamlessly into the post-study flow can reinforce consistency and help sustain long-term participation, particularly in peer-driven study environments.

Limitations and Future work

Our study revealed several facets of students' experiences in remote study rooms. However, this study had certain limitations. First, the small number of interview participants may limit the generalizability of our findings to a broader population of remote study room users. Additionally, because all participants were recruited in South Korea, our results might not reflect users' experiences from other countries or cultural backgrounds. Moreover, as all participants were university students in their twenties, our study may not capture the perspectives of older adults or individuals with disabilities, such as BLV or hard of hearing, who may encounter unique challenges when communicating online.

Future research should include a more diverse range of participants, including people from various regions, cultures, age groups, and individuals with disabilities. Broadening the participant base will increase the representativeness of the findings and provide valuable insights for designing more inclusive remote-group study tools. Additionally, while our study highlights how social presence and peer familiarity foster engagement and accountability, future research could explore ways to bring similar benefits to groups of unfamiliar participants. In particular, lightweight self-disclosure mechanismssuch as sharing names or study goalsmight help promote group cohesion and responsibility. Designing such features would require careful consideration of users' privacy preferences and their level of control over what is shared.

Conclusion

This study employed exploratory qualitative interviews were conducted to investigate the students' experiences using remote study rooms. We conducted semi-structured interviews with 13 students to examine their motivations for joining remote study sessions and their experiences before, during, and after the sessions. Our findings indicate that students actively motivated and supported one another throughout the entire study session. We also identified several key challenges encountered by the students during remote studying. Based on these insights, we propose design implications for technological solutions to mitigate

these challenges. We hope that our lifecycle-oriented approach encourages further research within the CSCW community on the motivations and inclusiveness of remote study environments.

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